Week 3

Class Activity 5

Group Activity: Differential Privacy in Model Training

Scenario: A hospital's data science team is building an AI model to predict employee attrition risk. They want to make sure their system follows Differential Privacy (DP) principles, meaning no individual employee's data should noticeably affect the overall result.

The model outputs the average attrition probability after each dataset update:

Dataset Version	Who's in the dataset	Model Output (Average Attrition Probability)
D_0	No private data yet	0.35
D_1	+ Alice's record	0.38
D_2	+ Alice + Jess's records	0.42

Your task is to help the team quantify the privacy loss (ε) as more people are added.

Step 1: Compute Privacy Loss

Step 2: Discussion Points

1. **Calculate each ε value and the total privacy loss after Alice and Jess are added.**
2. **Which dataset (D_0, D_1, or D_2) is most private?**
3. **What happens to privacy when more people's data is added without adding noise?**
4. **If the team wanted to maintain stronger privacy, what could they do (e.g., add noise, limit outputs, or cap ε)?**